What Does a Tuned System's Benchmark Score Mean? Optimism, Wrong Conclusions, and Reliable Reporting after Automatic Prompt and Pipeline Search

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Abstract

Automatic prompt and pipeline optimisers choose a configuration on a benchmark, and the winner's score on those same items is then reported as the system's performance. We ask what that number means and how to report one that means something. For exhaustive search over B candidates we derive the expected optimism, $\sigma^2(\tau^2 + \sigma^2)^{-1/2}\,\mathbb{E}Z_{(B)}$ (σ: evaluation noise, τ: true spread of the candidates), which grows with the search budget, shrinks with benchmark size and is largest when candidates truly differ least; it matches simulation ($r = 0.9975$ over 375 settings). We audit this on a real LLM and benchmark: Qwen2.5-1.5B-Instruct on MMLU-Pro with 72 prompt programs, replaying random search, successive halving, coordinate ascent and a MIPRO-like surrogate search on random benchmarks of 50–800 questions. With n=100, random search over all 72 reports $+5.4$ points more than the winner's true accuracy, and 83% of the reported improvement over an average configuration is optimism; a paired test declares the tuned system significantly better than an untuned default in 53% of benchmark draws although its true gain is 1.1 pt. Comparing reporting methods by known-truth bias, interval coverage, a no-signal control and false-win rates, we find that hold-out reporting is unbiased but costs precision, cross-fitting under-covers, Ladder and Thresholdout do not repair the bias at these sizes, and a replay bootstrap that re-runs the whole search on resamples removes 70–98% of the optimism with near-nominal intervals and the lowest error of the generic methods; for exhaustive search, empirical-Bayes shrinkage and a worst-case correction (subtract $\hat{\sigma}\,\mathbb{E}Z_{(B)}$) are about as accurate on the real pool, but shrinkage fails when a few candidates are much better. A simulator matched to the real pool reproduces the real audit ($r = 0.99$). Limits: one small model and benchmark, our own reimplementations of the search families (DSPy was not run end-to-end), and a real pool with little true spread.

1 Introduction

Automatic prompt and program optimisation is now a routine step in building LLM systems: a tool such as DSPy/MIPRO [Khattab et al., 2023, Opsahl-Ong et al., 2024] or an instruction-search procedure [Zhou et al., 2023] evaluates many candidate configurations (instructions, demonstrations, formats, module wirings) on the items of a benchmark and returns the best one. The number that then goes into the paper, the model card or the product decision is, almost always, the winner's score on *those same items*. That number is a maximum over noisy estimates. Even if every candidate were exactly as good as every other, the best of B of them would look better than average by roughly

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$\sigma\sqrt{2\ln B}$, where σ is the evaluation noise of a single candidate on n items; and decisions made from such numbers — "our tuned pipeline beats the baseline", "method A beats method B" — inherit the bias, particularly when the two sides were tuned with different effort.

This paper asks what a reported score means after automatic tuning, and how to report one that means something. We make four contributions.

1. **A quantitative account of the gap.** For exhaustive search over B candidates we give a closed form for the expected optimism, $\sigma^2(\tau^2+\sigma^2)^{-1/2}\,\mathbb{E}Z_{(B)}$, which makes explicit that it grows with the search budget, shrinks with benchmark size, and is *largest when the candidates truly differ least*, i.e. when tuning is least useful (Sec. 3); we verify it in 375 simulated settings.
2. **An audit on a real LLM and benchmark.** We cache the per-item correctness of 72 prompt programs (answer format $\times$ demonstration set $\times$ instruction $\times$ answer cue) for Qwen2.5-1.5B-Instruct on 1000 MMLU-Pro questions and replay four families of search procedures (random search, successive halving, coordinate ascent, a surrogate-guided MIPRO-like search) on random benchmark subsets of 50–800 items, scoring each selected configuration by its exact accuracy over the whole cached population (Sec. 4).
3. **Consequences for conclusions.** We measure how often a tuned system is declared significantly better than an untuned or lightly tuned competitor when it is not.
4. **Reporting methods and how to test them.** We compare the status quo with hold-out reporting, cross-fitting, a replay bootstrap that re-runs the *whole search* on resamples, empirical-Bayes shrinkage, conditional-on-selection inference and simultaneous intervals, and Ladder / Thresholdout, using a reliability protocol: known-truth coverage, a no-signal control, robustness to the shape of the candidate distribution, and decision error rates (Sec. 4).

2 Related Work

Selection bias and adaptive reuse of evaluation data. That the best of many models looks better on the data used to pick it than on new data is the classic selection-bias problem in model assessment [Cawley and Talbot, 2010]; bootstrap corrections of such optimism [Efron and Tibshirani, 1997] and nested resampling are standard in the statistical-learning literature. In machine learning competitions and benchmarks, adaptive reuse of one test set has been studied through the lens of adaptive data analysis: the reusable holdout (Thresholdout) [Dwork et al., 2015a,b] and the Ladder mechanism [Blum and Hardt, 2015] limit what a sequence of queries can reveal about the data, and large-scale replication studies found that benchmark accuracies hold up better than feared under test-set reuse but fall under distribution shift [Recht et al., 2019]. Our setting differs in that the "analyst" is an automatic tuner whose queries (candidate prompts or programs) are numerous, adaptive, and issued without human restraint, and the reported object is the score of the single winning configuration.

Inference after selection. Econometrics and statistics offer estimators and intervals for the *winner* of a selection: conditional and hybrid inference [Andrews et al., 2024], polyhedral conditioning [Lee et al., 2016], valid post-selection inference [Berk et al., 2013], and Gaussian multiplier-bootstrap bounds for maxima [Chernozhukov et al., 2013]. We use these as ingredients and ask how they behave for binary per-item outcomes shared across configurations and for search algorithms (successive halving, surrogate-guided search) whose selection event is not a simple polyhedron.

Prompt and program optimisation, and evaluation noise. Automatic search over instructions and demonstrations is now routine [Zhou et al., 2023, Khattab et al., 2023, Opsahl-Ong et al., 2024], including budget-aware formulations as best-arm identification [Shi et al., 2024]. Prompt-sensitivity studies show that formatting and wording alone move benchmark scores by many points [Sclar et al., 2024, Mizrahi et al., 2024], and efficient multi-prompt evaluation estimates the whole performance distribution over templates [Polo et al., 2024]; those works treat the prompt as given, whereas we study the bias introduced when the prompt is *chosen* on the benchmark. Perez et al. [2021] pointed out that prior "few-shot" work tuned prompts and hyper-parameters on many held-out examples and studied model selection (cross-validation, minimum description length) when no such examples are available; Miller [2024] advocates error bars for evaluations, but treats the evaluated system as fixed.

3 Setting, Theory and Reporting Methods

Setting. A *system* is a configuration θ from a finite set Θ ($|\Theta| = K$) of prompts / few-shot demonstration sets / pipeline variants, scored per item by $y(\theta, x) \in \{0, 1\}$ (we use correctness). Its true performance is $\mu(\theta) = \mathbb{E}_{x\sim D}\, y(\theta, x)$ for a task distribution D. A *benchmark* is a sample S of n items. A *search procedure* $\mathcal{A}$ (random search, successive halving, coordinate ascent, surrogate-guided search, ...) interacts with S only by evaluating configurations on (subsets of) S and returns $\hat{\theta} = \mathcal{A}(S)$. The usual *reported score* is $\hat{R} = \frac{1}{n}\sum_{x\in S} y(\hat{\theta}, x)$. We call $\mathrm{opt} = \hat{R} - \mu(\hat{\theta})$ the *optimism* of the report; the natural quantities of interest are its mean (bias), whether intervals around $\hat{R}$ cover $\mu(\hat{\theta})$, and whether decisions ("A beats B") based on reported scores are correct.

A closed-form benchmark for the size of the effect. Let $\bar{Y}_c = \mu_c + \varepsilon_c$ be the accuracies of B fully evaluated configurations, with $\mu_c \sim \mathcal{N}(m, \tau^2)$ i.i.d. (true spread τ) and configuration-specific evaluation noise $\varepsilon_c \sim \mathcal{N}(0, \sigma^2)$ i.i.d. Then $\bar{Y}_{\hat{c}} = m + \sqrt{\tau^2 + \sigma^2}\, Z_{(B)}$, with $Z_{(B)}$ the maximum of B standard normals, and since $\mathbb{E}[\mu_c \mid \bar{Y}_c] = m + \kappa(\bar{Y}_c - m)$, $\kappa = \tau^2/(\tau^2 + \sigma^2)$,

$$\mathbb{E}[\hat{R} - \mu(\hat{\theta})] = \frac{\sigma^2}{\sqrt{\tau^2 + \sigma^2}}\,\mathbb{E}Z_{(B)}, \qquad \mathbb{E}[\mu(\hat{\theta}) - m] = \frac{\tau^2}{\sqrt{\tau^2 + \sigma^2}}\,\mathbb{E}Z_{(B)}. \tag{1}$$

Hence (i) the optimism grows with the search budget like $\mathbb{E}Z_{(B)} \approx \sqrt{2\ln B}$ and, because $\sigma^2 \propto 1/n$ (for binary scores $\sigma^2 \approx (1-\bar{r})\, p(1-p)/n$, where $\bar{r}$ is the average correlation between configurations' per-item outcomes: shared item difficulty cancels in comparisons and does not inflate the optimism), shrinks with n like $n^{-1/2}$ when $\tau \ll \sigma$ and like n^{-1} when $\tau \gg \sigma$; (ii) the share of the reported improvement over an average configuration that is noise is $\sigma^2/(\tau^2 + \sigma^2)$ — when configurations truly differ little compared with the evaluation noise ($\tau \ll \sigma$), *all* of the reported gain is optimism and the optimism is at its maximum $\sigma\,\mathbb{E}Z_{(B)}$; (iii) without the Gaussian-prior structure the maximal inequality gives the bound $\mathbb{E}[\hat{R} - \mu(\hat{\theta})] \le \sigma_{\max}\sqrt{2\ln B}$ for exhaustive search over B configurations. Equation (1) is an idealisation; Section 4 tests it, and the other regimes, by simulation.

Reporting methods compared. All methods see only S (size n) and the search procedure.

- **Naive** (status quo): $\hat{R}$ with a Wald interval.
- **Hold-out**: search on a random $(1-\rho)n$ items, report the accuracy of the winner on the remaining ρn with a Wilson interval; exact for $\mu(\hat{\theta}_1)$ but spends ρn items on reporting instead of search.
- **Cross-fit**: F-fold: search on $F-1$ folds, score on the held-out fold, pool. Estimates the performance of the *procedure*, using F extra searches.
- **Replay bootstrap** (generic, our default recommendation when the full score matrix of explored configurations is available or evaluations are cached): re-run the *entire* search $\mathcal{A}$ on $b = 1..100$ bootstrap resamples S^{*b} of S, set $\widehat{\mathrm{opt}} = \frac{1}{b}\sum_b [\hat{R}^{*b} - \hat{R}_S(\hat{\theta}^{*b})]$ where $\hat{R}^{*b}$ is the winner's score on S^{*b} and $\hat{R}_S(\hat{\theta}^{*b})$ its score on the original S, and report $\hat{R} - \widehat{\mathrm{opt}}$ (Efron-style optimism correction applied to the whole procedure). A `.632` variant uses out-of-bag items.
- **Empirical-Bayes shrinkage** (random/grid search over B fully evaluated candidates): estimate τ^2 from the between-configuration variance after subtracting the paired item-level noise, shrink the winner toward the grand mean by κ (eq. 1, the Tweedie/normal-normal correction).
- **Worst-case analytic correction** (random/grid search): subtract $\hat{\sigma}\,\mathbb{E}Z_{(B)}$, eq. (1) with τ=0 and $\hat{\sigma}$ the paired item-level noise of the configuration contrasts: the maximal optimism for given (n, B), needing no prior on candidate quality; conservative whenever candidates truly differ.
- **Conditional (winner) inference** (random/grid search): the polyhedral-lemma truncated-normal interval and median-unbiased estimate for $\mu_{\hat{c}}$ given that $\hat{c}$ won, with the covariance of the B accuracies estimated from the paired item-level data [Lee et al., 2016, Andrews et al., 2024].
- **Simultaneous interval**: max-t bootstrap band over the B evaluated configurations [Chernozhukov et al., 2013]; valid for any selection rule among them.

- **Ladder** [Blum and Hardt, 2015] and **Thresholdout** [Dwork et al., 2015a], run over the same candidate sequence, as adaptive-data-analysis baselines.

4 Experiments

4.1 Experimental setup

Real LLM and benchmark. *Data.* MMLU-Pro [Wang et al., 2024] (dataset revision `b189ec7`, 12 032 test questions, up to 10 options). We fix a uniformly random population P of 1000 test questions; the remaining questions are never used. The 70 validation questions provide few-shot demonstrations. *Model.* Qwen2.5-1.5B-Instruct [Qwen Team, 2024], 8-bit GGUF, run with llama.cpp on CPU; the answer is the option letter with the largest next-token log-probability (greedy, deterministic, no chain of thought), so a (configuration, question) pair has one fixed outcome. *Configuration space* (K=72): option format (`A. text` / `(A) text`) $\times$ demonstrations (none / one of two fixed 3-shot sets) $\times$ one of 4 instruction sentences $\times$ one of 3 answer cues. This is a small, hand-built space of the kind that instruction/demonstration optimisers explore, evaluated exhaustively so that every search is a replay on a complete per-item score matrix $Y \in \{0,1\}^{|P|\times 72}$. *Benchmark draws and truth.* Each replicate draws a benchmark S of $n \in \{50, 100, 200, 400, 800\}$ questions i.i.d. (with replacement) from P, and the true performance of a configuration is its exact accuracy over P: the empirical distribution of P plays the role of the task distribution. (Drawing S without replacement and scoring on $P \setminus S$ is also possible, but then the fresh accuracy is mechanically anti-correlated with the benchmark accuracy and the measured optimism is inflated by $|P|/(|P| - n)$; App. A.6 reports this secondary design for small n.) *Search procedures* (each sees only S): random search with budget $B \in \{1, \ldots, 72\}$ and full evaluation of each candidate; successive halving (3-way, nested subsets of S, final rung on all of S); greedy coordinate ascent over the three factors from a random start; and a surrogate-guided search in the style of MIPRO [Opsahl-Ong et al., 2024]: 40 trials on random minibatches of 50 questions proposed from an additive per-factor surrogate, then full evaluation of the 3 best. These are our own implementations of the algorithm families, *not* the DSPy code (Sec. 5).

Synthetic worlds. To control the quantities that theory says matter and to reach regimes the real pool does not contain, we simulate item-level outcomes from a latent-threshold model with the same factorial configuration structure: $y_{ci} = \mathbb{1}[a + \mu_c + \sqrt{\rho}\, z_i + \sqrt{1-\rho}\, u_{ci} > 0]$, $z_i, u_{ci} \sim \mathcal{N}(0,1)$, where μ_c (standardised to SD τ) mixes additive factor effects and interactions, ρ is the latent between-configuration correlation (item difficulty shared by all configurations), and a sets base accuracy. We also use i.i.d. configuration effects with normal, sparse (5% of configurations much better), heavy-tailed (t_3) and bimodal (half the configurations "broken") distributions for robustness checks.

Metrics. For each reporting method: *bias* $= \mathbb{E}[\text{report} - \mu(\hat{\theta})]$, *RMSE*, *coverage* and median width of nominal-95% intervals for $\mu(\hat{\theta})$ (the quality of the configuration that would actually be deployed), and, for comparisons, the rate of significant "$A > B$" claims at one-sided level 5% against the true gap. Monte-Carlo errors are reported (bias: ± 1.96 SE; coverage: binomial SE).

4.2 How large is the gap, and what drives it?

Real pool. The 72 prompt programs are nearly interchangeable: on the 1000 cached questions their accuracies span 25.8–31.0% (SD 1.2 pts, of which about 0.8 pts is evaluation noise on $|P| = 1000$ items; the noise-corrected true spread is $\hat{\tau} \approx 0.9$ pt) and the mean pairwise correlation of their per-question outcomes is 0.66. Tuning by random search on n=100 questions reports an accuracy +5.4 pts above the winner's true accuracy (95% MC interval ± 0.4; Fig. 1a); at n=50 the figure is +7.8, at n=800 still +1.2. The reported improvement over an average configuration at n=100 is 6.7 pts, the true one 1.1 pts: 83% of the reported gain is optimism (Fig. 1b). The other search procedures behave alike (Table 2, App.): successive halving and the minibatch-driven surrogate search spend fewer full evaluations and over-report less (+3.2 and +3.7 pts at n=100), coordinate ascent lies in between (+4.4).

Drivers. Equation (1) says that optimism $= c(n)\, \mathbb{E}Z_{(B)}$ with $c = \sigma^2/\sqrt{\tau^2 + \sigma^2}$, and the real pool follows it closely. *Search budget:* dividing the measured optimism of random search by $\mathbb{E}Z_{(B)}$ gives a constant across $B \in \{2, \ldots, 72\}$ (at n=100: 2.3 pts, SD over B 0.06); even B=8

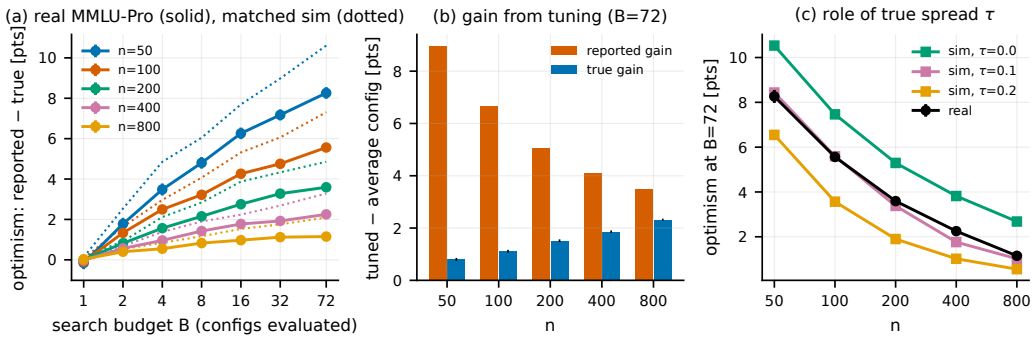


Figure 1: **Status-quo optimism on a real LLM/benchmark.** (a) Reported minus true accuracy of the winner of random search over B configurations (Qwen2.5-1.5B, MMLU-Pro; solid, ± 1.96 SE) and the same quantity in a synthetic world matched to the real pool's accuracy level, spread and correlation (dotted). (b) Improvement of the tuned configuration over the average configuration: reported vs. true. (c) Optimism at B=72 in synthetic worlds with true configuration spread τ (probit units) and on the real pool.

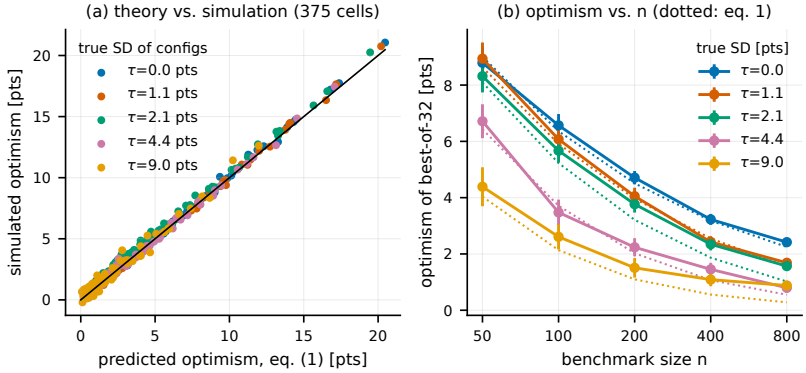


Figure 2: **Closed-form optimism (eq. 1) vs. simulation.** (a) 375 settings with i.i.d. Gaussian configuration effects (5 spreads $\times$ 3 correlations $\times$ 5 n $\times$ 5 B). (b) Dependence on n for best-of-32 (ρ=0.8); dotted = formula.

gives +3.3 pts. *Benchmark size:* $c(n)$ measured on the real pool is 3.4/2.3/1.5/1.0/0.6 pts for n=50/100/200/400/800; the formula, fed only the pool's measured evaluation noise (config-specific part of the per-item variance, $\sigma^2 \propto 1/n$) and noise-corrected spread $\hat{\tau}$, predicts 3.6/2.5/1.7/1.1/0.7, within about 10%. It falls between n^{-1} and $n^{-1/2}$, and $n \approx 800$ is needed before best-of-72 drops to about one point. *True spread:* optimism is *largest when configurations truly differ least* (Fig. 1c): τ=0 is the worst case, and a pool whose true SD is 0.2 probit units ($\approx$ 7 pts) roughly halves it at n=100. Over 375 simulated settings (n, B, τ, between-configuration correlation) predicted and simulated optimism correlate at $r = 0.9975$ (mean absolute error 0.25 pts; Fig. 2).

4.3 Does the gap change conclusions?

Table 1 shows what this does to claims. On the real pool, tuning with B=72 gives a true gain of only 1.1 pts over a random default at n=100, yet the naive paired test declares the tuned system significantly better in 53% of benchmark draws (and in 22% of draws the true gain is below one point). Against a competitor that also tuned, but with budget 4, the naive gap is 3.9 pts against a true gap of 0.6; 22% of draws yield a significant win. At n=400 the problem persists (64% claims against a 1.8-pt true gain). Hold-out reporting restores nominal behaviour (8% claims, 4% false at n=100) at the price of power in a world with a real effect (simulated τ=0.1, n=100, true gap 5.8 pts: 23% of draws declare a win with hold-out reporting, 43% with the replay-bootstrap gap, 80% naively),

Table 1: **Decisions based on reported scores.** System A = random search with B=72 on n questions. It is compared on the same questions with an untuned default (a random configuration) or with a lightly tuned system (RS, B=4) by a paired one-sided z-test at 5%. *Claims*: share of benchmark draws declaring A significantly better; *false claims*: such declarations when the true gap is below 1 pt. True gap = accuracy difference of the deployed configurations over the population. 300 draws per row (real) / 150 (simulated).

	true gap [pts]	naive gap [pts]	claims / false claims [%] naive	replay-boot.	hold-out
real, vs. untuned default, n=100	1.1	6.8	53 / 22	15 / 6	8 / 4
real, vs. untuned default, n=400	1.8	3.9	64 / 12	24 / 3	11 / 2
real, vs. RS B=4, n=100	0.6	3.9	22 / 13	5 / 3	8 / 4
real, vs. RS B=4, n=400	1.0	2.2	24 / 8	11 / 4	10 / 2
sim, τ=0, vs. untuned default, n=100	0.0	7.4	48 / 48	11 / 11	6 / 6
sim, τ=0, vs. untuned default, n=400	0.0	3.5	52 / 52	5 / 5	7 / 7
sim, τ=0, vs. RS B=4, n=100	0.0	4.0	9 / 9	3 / 3	9 / 9
sim, τ=0, vs. RS B=4, n=400	0.0	2.2	9 / 9	1 / 1	1 / 1
sim, τ=0.1, vs. untuned default, n=100	5.8	11.2	80 / 4	43 / 0	23 / 1
sim, τ=0.1, vs. untuned default, n=400	7.1	8.9	88 / 0	82 / 0	47 / 0
sim, τ=0.1, vs. RS B=4, n=100	2.7	6.6	38 / 5	23 / 2	15 / 2
sim, τ=0.1, vs. RS B=4, n=400	3.4	4.5	63 / 1	49 / 0	21 / 1

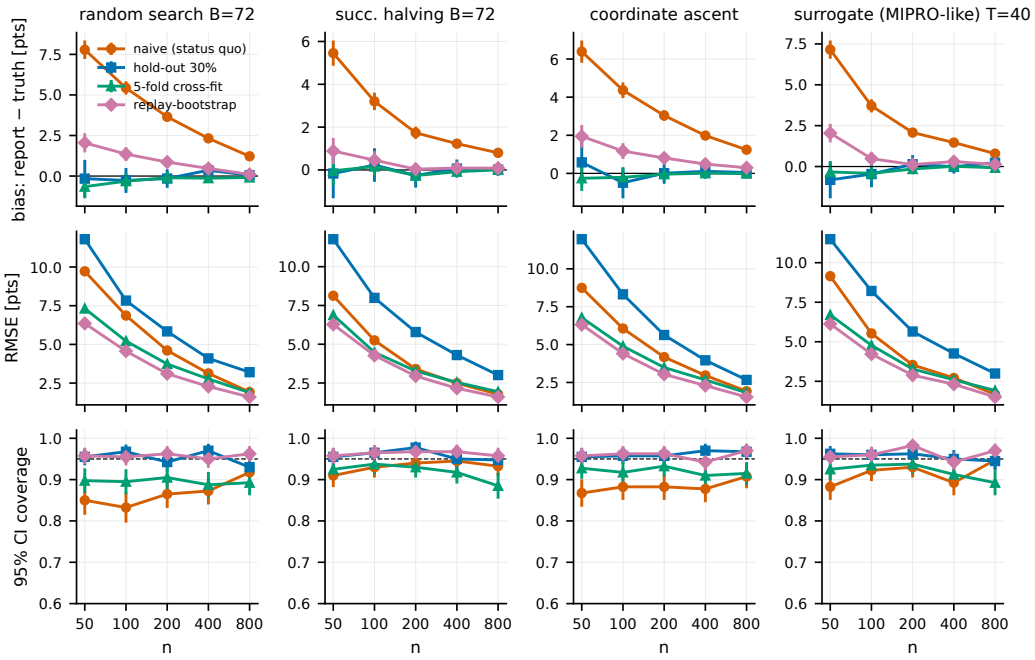


Figure 3: **Reporting methods on the real pool.** Bias, RMSE and coverage of nominal-95% intervals for the accuracy of the configuration that is deployed, for four search procedures and n from 50 to 800 (400 benchmark draws per cell; bars $\pm$1.96 SE). Synthetic counterparts: Fig. 4 (App.).

while with no true spread (τ=0) the naive test declares a win in 48% of draws against 5% nominal (replay-bootstrap 11%, hold-out 6%).

4.4 Which reporting method is reliable?

Figure 3 summarises the audit; Tables 3 and 4 (App.) give numbers. *Status quo.* Biased upward everywhere; its 95% intervals cover the winner's true accuracy only 83% of the time (RS, n=100; range over cells 83–95%). *Hold-out* (30% of S reserved) is unbiased in every cell ($|\text{bias}| \leq 0.8$ pts) and its intervals are valid (93–98%), but its error is the largest: RMSE 7.8 vs. 6.9 pts for the naive

number at $n{=}100$, because 30% of the data are removed from the reporting estimate *and* the search sees fewer items. *Cross-fitting* (5 folds) is nearly unbiased and has lower RMSE than hold-out, but its intervals undercover (88–94%) because they ignore the variability of the selection. *Replay bootstrap* (100 replays; 10 already suffice, App. A.4) removes most of the bias (RS $n{=}100$: +5.4 $\rightarrow$ +1.4 pts; for SH and MIPRO-like search the residual is below 1 pt at $n \geq 100$ in the real pool), has the lowest RMSE of all generic methods (4.6 pts vs. 6.9) and covers at 94–98% (nominal 95) with the same interval width as the naive one. It under-corrects when the true spread is nil and the budget large (App., Table 3): a positive residual of 1–2.5 pts. *Empirical-Bayes shrinkage* (applicable when B candidates are all evaluated on S) is best in the real pool (RS $B{=}72$, $n{=}100$: bias +0.0, RMSE 4.0 pts, intervals $0.80\times$ as wide as the naive ones with 94% coverage) but it relies on a normal prior for configuration quality: in the synthetic robustness check (Table 5) it over-shrinks a sparse set of genuinely better configurations (bias -4.1 pts, coverage 74%). The worst-case correction (subtract $\hat{\sigma}\,\mathbb{E}Z_{(B)}$) is as accurate on the real pool (bias -0.8, RMSE 4.0 pts, coverage 98%) and degrades more gracefully (sparse pool: -3.3 pts, heavy-tailed: -1.5) because it errs on the conservative side by construction. The conditional (winner) intervals are valid in all worlds but are $1.6\times$ as wide as the naive ones and unbounded in a sizeable fraction of draws; simultaneous intervals are also valid but, being a statement about all candidates, do nothing about the point estimate. Ladder reduces the bias only partially (RS: +1.4 pts, coverage 91%) and Thresholdout with a half-split of S and noisy answers is *worse* than doing nothing at these budgets (+9.9 pts), as its guarantees need far more data per query.

4.5 Further diagnostics

Demonstrations drawn from the benchmark leak their own answers. Optimisers such as DSPy's `BootstrapFewShot` take demonstrations from the tuning set, so the tuned prompt contains the answer to some of the items on which it is then scored. With the same prompt format as above, accuracy on the three items shown as demonstrations is 75% versus 30% on other items (40 random demonstration sets), i.e. about $+45$ pts per leaked item: 3-shot bootstrapped demonstrations add roughly 1.4 pts to a report at $n{=}100$ on top of the selection effect (arithmetic from these two numbers; not an end-to-end DSPy run).

A label-permutation negative control. Re-scoring the cached predictions against permuted gold labels (all configurations then truly at chance) and re-running the search measures what the procedure can fabricate from noise alone. Its optimism is a fraction 0.64–1.01 (median 0.82) of the optimism measured with real labels (App.), a cheap audit of how much of a reported gain a search of that size and benchmark size could produce spuriously; it is not an unbiased estimator because the noise level depends on accuracy.

How much data should reporting get? For random search with $B{=}72$ and $n{=}100$ on the real pool, reserving 10%/30%/50% of the benchmark for reporting gives hold-out RMSE 13.9/8.1/6.6 pts (naive: 6.9; replay bootstrap: 4.6), while the true accuracy of the configuration deployed falls by only 0.04/0.08/0.30 pts, because searching with fewer items hardly matters when the candidates are nearly equivalent. In a simulated world with real spread ($\tau{=}0.1$) the same loss is 0.17/0.49/0.95 pts. The price of an unbiased report is thus mainly precision of the report, not the quality of the system; in small benchmarks the replay bootstrap offers a better error trade-off.

A pool with real signal (Real-II). To check that the picture is not specific to a pool of near-equivalent prompts, we built a second real pool from four small models (Qwen2.5-1.5B, Qwen2-1.5B, Qwen3-1.7B, Qwen3-0.6B) $\times$ 8 prompt variants (zero-/3-shot, 2 instructions, 2 cues), i.e. $K{=}32$ configurations scored on 250 questions (smaller because of compute; App. A.9). Here the pool's accuracies span 17.2–28.8% (SD 3.3 pts), driven mostly by model identity. Random search over all 32 on $n{=}100$ questions still over-reports by +4.8 pts, presumably because the final choice is among near-equivalent prompts of the best model(s); replay bootstrap and hold-out give +0.8 and -0.4 (coverage 93% and 97% vs. 88% naive). Larger spread across models does not make the reported number trustworthy when the final choice is made among close competitors.

5 Discussion, Recommendations and Limitations

What a reported score means. After automatic search, the winner's score on the search set is a maximum over noisy estimates. In our real audit it overstated the winner's true accuracy by 1–8 pts depending on n, budget and search procedure, and more than four fifths of the reported improvement over an average configuration at $n{=}100$ was optimism, because the 72 programs were (nearly) equally good. The effect is not a property of one tool: it appears for exhaustive random search, bandit-style successive halving, greedy coordinate ascent and a surrogate-guided search alike, and it is governed by a simple ratio of evaluation noise to true spread (eq. 1). Whether a score can be taken at face value therefore depends on (n, B, τ), which a reader can check only if the paper states the search budget and benchmark size; when τ is large relative to $n^{-1/2}$ the naive number is nearly right (Fig. 1c), when it is small it is mostly noise.

Recommendations (supported by the experiments above). (i) State n, the number of configurations evaluated and the search procedure, and label the search-set score as such. (ii) Whenever a few hundred items can be spared, report the score of the selected configuration on items never used by the search; it is unbiased and has valid intervals, but plan for its cost: reserving 30% of the benchmark gave an RMSE of 7.8 pts at $n{=}100$ against 4.6 for the replay bootstrap. (iii) If all search-time evaluations are cached (or can be replayed cheaply), additionally run the *replay bootstrap*: it removed 70–98% of the optimism (median 79%) and had the lowest error among generic methods, with near-nominal intervals; where it under-corrects (zero true spread, large budget) it errs on the conservative side of coverage but still leaves 1–2.5 pts. (iv) For exhaustive/random search over a pool with no outliers, empirical-Bayes shrinkage is more accurate still, but check the shape of the pool: it fails when a few configurations are genuinely much better. (v) For comparisons, apply the same protocol to both sides (tuned-vs-untuned and unequal budgets are the dangerous cases) and base the test on the corrected or held-out gap. (vi) Test any new reporting method as we did: known-truth bias and coverage over many benchmark draws, a no-signal control (true spread zero; here also a label-permutation control), robustness to the shape of the candidate distribution, and the rate of false "wins".

Limitations. *Scale and scope.* The real audit uses one small model (Qwen2.5-1.5B, 8-bit, CPU), one benchmark (MMLU-Pro), direct next-token scoring (deterministic; no generation, no chain-of-thought, no sampling noise, no LLM-judged metrics) and a small hand-built configuration space whose members turned out to be nearly equivalent (true spread ≈ 0.9 pt). Regimes with large true spread, multi-stage programs and larger models are covered only by simulation and by the theory; the simulated worlds are tuned to be neither adversarial nor exhaustive. The compute budget (4 CPU cores, no GPU) ruled out larger models and per-method LLM calls. *Search procedures are our reimplementations of algorithm families* (random search, successive halving, coordinate ascent, a MIPRO-like surrogate search), replayed on a cached score matrix; we did *not* run DSPy/MIPRO end-to-end. A smoke test of DSPy 3.4.1 with a local 0.5B model produced unparsable outputs and $\sim$45 s per request on one core, so we abandoned it; the demonstration-leak measurement (Sec. 4) is an isolated check, not an end-to-end replication. *Truth.* The population is the 1000 cached questions; benchmarks are drawn i.i.d. from this empirical distribution and truth is the exact accuracy on it. This avoids a finite-pool artefact (App. A.6) but implies duplicate questions in benchmarks whose size approaches $|P|$ ($n{=}800$ draws contain about 55% distinct questions) and means "true" refers to this distribution; confidence in its relation to MMLU-Pro as a whole is limited by $|P|$. Fresh questions are i.i.d. with the benchmark; we do not study distribution shift. *Replay bootstrap* needs per-item scores of everything the replayed search touches; for adaptive procedures that generate new candidates (LLM-proposed instructions) the replay must be restricted to candidates already evaluated, a variant we did not test. It under-corrects at small spread/large budget, and its cost is b extra searches (free on a cache). *Baselines.* Ladder and Thresholdout were run with one or three untuned parameter settings and are known to need more data than we gave them; our results say they are not drop-in fixes at these sizes, not that they are useless. *Hyper-parameters* of our own methods (30% hold-out, 100 bootstrap replicates, 5 folds, normal prior for EB) were fixed from synthetic pilots before the real data were analysed and were not tuned on the real pool.

References

Isaiah Andrews, Toru Kitagawa, and Adam McCloskey. Inference on winners. *The Quarterly Journal of Economics*, 139(1):305–358, 2024.

Richard Berk, Lawrence Brown, Andreas Buja, Kai Zhang, and Linda Zhao. Valid post-selection inference. *The Annals of Statistics*, 41(2):802–837, 2013.

Avrim Blum and Moritz Hardt. The ladder: A reliable leaderboard for machine learning competitions. In *International Conference on Machine Learning*, 2015.

Gavin C. Cawley and Nicola L. C. Talbot. On over-fitting in model selection and subsequent selection bias in performance evaluation. *Journal of Machine Learning Research*, 11:2079–2107, 2010.

Victor Chernozhukov, Denis Chetverikov, and Kengo Kato. Gaussian approximations and multiplier bootstrap for maxima of sums of high-dimensional random vectors. *The Annals of Statistics*, 41(6): 2786–2819, 2013.

Cynthia Dwork, Vitaly Feldman, Moritz Hardt, Toniann Pitassi, Omer Reingold, and Aaron Roth. Generalization in adaptive data analysis and holdout reuse. In *Advances in Neural Information Processing Systems*, 2015a.

Cynthia Dwork, Vitaly Feldman, Moritz Hardt, Toniann Pitassi, Omer Reingold, and Aaron Roth. Preserving statistical validity in adaptive data analysis. In *Proceedings of the 47th Annual ACM Symposium on Theory of Computing (STOC)*, 2015b.

Bradley Efron and Robert Tibshirani. Improvements on cross-validation: the .632+ bootstrap method. *Journal of the American Statistical Association*, 92(438):548–560, 1997.

Omar Khattab, Arnav Singhvi, Paridhi Maheshwari, Zhiyuan Zhang, Keshav Santhanam, Sri Vardhamanan, Saiful Haq, Ashutosh Sharma, Thomas T. Joshi, Hanna Moazam, Heather Miller, Matei Zaharia, and Christopher Potts. DSPy: Compiling declarative language model calls into self-improving pipelines. *arXiv preprint arXiv:2310.03714*, 2023.

Jason D. Lee, Dennis L. Sun, Yuekai Sun, and Jonathan E. Taylor. Exact post-selection inference, with application to the lasso. *The Annals of Statistics*, 44(3):907–927, 2016.

Evan Miller. Adding error bars to evals: A statistical approach to language model evaluations. *arXiv preprint arXiv:2411.00640*, 2024.

Moran Mizrahi, Guy Kaplan, Dan Malkin, Rotem Dror, Dafna Shahaf, and Gabriel Stanovsky. State of what art? a call for multi-prompt LLM evaluation. *Transactions of the Association for Computational Linguistics*, 12, 2024.

Krista Opsahl-Ong, Michael J. Ryan, Josh Purtell, David Broman, Christopher Potts, Matei Zaharia, and Omar Khattab. Optimizing instructions and demonstrations for multi-stage language model programs. In *Proceedings of EMNLP*, 2024.

Ethan Perez, Douwe Kiela, and Kyunghyun Cho. True few-shot learning with language models. In *Advances in Neural Information Processing Systems*, 2021.

Felipe Maia Polo, Ronald Xu, Lucas Weber, Mírian Silva, Onkar Bhardwaj, Leshem Choshen, Allysson Flavio Melo de Oliveira, Yuekai Sun, and Mikhail Yurochkin. Efficient multi-prompt evaluation of LLMs. In *Advances in Neural Information Processing Systems*, 2024.

Qwen Team. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.

Benjamin Recht, Rebecca Roelofs, Ludwig Schmidt, and Vaishaal Shankar. Do ImageNet classifiers generalize to ImageNet? In *International Conference on Machine Learning*, 2019.

Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. Quantifying language models' sensitivity to spurious features in prompt design or: How I learned to start worrying about prompt formatting. In *International Conference on Learning Representations*, 2024.

Chengshuai Shi, Kun Yang, Zihan Chen, Jundong Li, Jing Yang, and Cong Shen. Efficient prompt optimization through the lens of best arm identification. In *Advances in Neural Information Processing Systems*, 2024.

Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, Abhranil Chandra, Shiguang Guo, Weiming Ren, Aaran Arulraj, Xuan He, Ziyan Jiang, et al. MMLU-Pro: A more robust and challenging multi-task language understanding benchmark. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, 2024.

Yongchao Zhou, Andrei Ioan Muresanu, Ziwen Han, Keiran Paster, Silviu Pitis, Harris Chan, and Jimmy Ba. Large language models are human-level prompt engineers. In *International Conference on Learning Representations*, 2023.

A Additional results and details

A.1 Search procedures and reporting-method details

Random search (RS, budget B): B distinct configurations drawn uniformly from the pool, all evaluated on all of S, argmax (random tie-break). *Successive halving*: B random candidates, reduction factor 3, rungs evaluated on nested random subsets of S of size $n/3^r$ (at least 25), the last rung on all of S, winner = best on the last rung. *Coordinate ascent*: random start, then up to two passes over the three factors (format×demonstrations, instruction, cue) in random order, each time evaluating all levels of one factor on all of S and moving to the best (about 26 evaluations). *MIPRO-like*: 40 trials; a trial samples each factor level either uniformly (probability 0.3 or first two trials) or from a softmax (temperature 0.03) over the running mean minibatch score of the levels, evaluates the sampled configuration on a random minibatch of 50 items of S, then the 3 configurations with the highest average minibatch score are evaluated on all of S and the best is returned. Reports are always the winner's score on all of S. *Hold-out*: random 30% of S held out; search on the remaining 70%. *Cross-fit*: 5 random folds; interval = Wald on pooled held-out outcomes. *Replay bootstrap*: 100 resamples (with replacement) of S, the whole search (including its internal randomness) re-run on each; interval = corrected point estimate $\pm 1.96\sqrt{\hat{p}(1-\hat{p})/n}$. *EB*: interval = estimate $\pm 1.96\sqrt{\kappa\hat{\sigma}^2 + \hat{s}_m^2}$, $\hat{s}_m^2$ the sampling variance of the grand mean over items (accounts for shared item noise). *Conditional*: polyhedral truncated-normal CI inverted by root-finding, covariance from the item-level matrix. *Simultaneous*: 500 bootstrap draws of the max-t statistic. *Ladder*: step $\eta = n^{-1/2}$, reported = rounded best. *Thresholdout*: S split in halves; threshold T and Laplace noise scale σ_T as multiples of the SE of the difference of half-sample accuracies (main setting T=1·SE, σ_T=0.1·SE; two other settings were also tried and were no better).

A.2 Optimism of the status-quo report on the real pool

Table 2: Reported minus true accuracy of the selected configuration (pts, ± 1.96 MC SE), real pool, benchmark of n questions drawn i.i.d. from the cached population; 400 draws per cell.

	n=50	100	200	400	800
RS B=8	5.0±0.6	3.3±0.4	2.2±0.3	1.3±0.2	0.8±0.2
RS B=72	7.8±0.6	5.4±0.4	3.7±0.3	2.3±0.2	1.2±0.1
SH B=72	5.5±0.6	3.2±0.4	1.7±0.3	1.2±0.2	0.8±0.2
coord. asc.	6.4±0.6	4.4±0.4	3.0±0.3	2.0±0.2	1.2±0.1
MIPRO-like	7.2±0.6	3.7±0.4	2.1±0.3	1.5±0.2	0.8±0.1

Table 3: Synthetic worlds (ρ=0.8; τ in probit units; 300 draws per cell). Entries: bias / RMSE (pts) / coverage (%) for the accuracy of the deployed configuration.

setting	naive	hold-out	cross-fit	replay-boot.
τ=0.0, RS B=72, n=100	+7.6/8.5/76	+0.2/9.5/91	-0.0/5.7/92	+2.5/4.8/95
τ=0.0, RS B=72, n=400	+3.8/4.3/65	-0.3/4.3/94	-0.2/2.8/92	+1.1/2.3/99
τ=0.0, MIPRO-like, n=100	+5.6/7.0/86	-0.2/8.6/96	+0.2/4.7/97	+1.6/4.6/97
τ=0.0, MIPRO-like, n=400	+2.1/3.0/88	+0.5/4.3/94	+0.4/2.4/96	+0.6/2.2/97
τ=0.1, RS B=72, n=100	+5.6/7.4/80	+0.5/9.1/95	+0.1/6.6/82	+1.4/5.3/94
τ=0.1, RS B=72, n=400	+1.7/2.7/95	-0.1/4.7/94	-0.2/2.9/94	+0.1/2.3/97
τ=0.1, MIPRO-like, n=100	+4.4/6.3/88	+1.1/8.6/96	+0.1/5.8/91	+1.0/4.8/96
τ=0.1, MIPRO-like, n=400	+1.1/2.5/95	+0.2/4.1/96	+0.1/3.3/87	+0.0/2.3/97

Table 4: Real pool, random search B=72 evaluated exhaustively on S: bias / RMSE / coverage (%) / median interval width (pts). Conditional: median-unbiased estimator is unbounded in a sizeable fraction of draws, so only coverage and width are given.

	naive	EB shrink	worst-case corr.	conditional	simultaneous	Ladder	Thresholdout
RS B=72, n=100	+5.4/6.8/86/19	+0.0/4.0/94/15	-0.8/4.0/98/19	–/–/97/30	+5.4/6.8/98/28	+1.4/4.8/91/20	+9.9/11.0/78/28
RS B=72, n=400	+2.1/3.0/89/9	-0.3/2.2/90/8	-1.0/2.4/94/9	–/–/97/15	+2.1/3.0/99/14	+1.0/2.6/94/10	+4.6/5.2/86/14

A.3 Reporting methods: numbers

A.4 Number of bootstrap replays

A.5 Label-permutation negative control

A.6 Why the primary real analysis draws benchmarks with replacement

If S is drawn without replacement from a finite pool P of N questions and truth is taken as the accuracy on the remaining $N-n$ questions, then for any fixed configuration $\mathrm{fresh} = \bar{y}_P - \frac{n}{N-n}(\bar{y}_S - \bar{y}_P)$: the fresh accuracy is mechanically anti-correlated with the benchmark accuracy, and the measured optimism is inflated by the factor $N/(N-n)$ relative to its value for a benchmark drawn from the population P itself. With $N = 1000$ the factor is 1.11 at n=100 and 1.67 at n=400. We therefore draw S i.i.d. from P (with replacement) and use the exact accuracy over P as truth. As a check, for $n \leq 200$ where the inflation is small, benchmarks drawn without replacement and scored on $P \setminus S$ give naive / replay-bootstrap / hold-out biases of +8.5/+2.8/+0.1 (n=50), +6.1/+2.0/+0.1 (n=100) and +4.1/+1.3/-0.1 (n=200) pts for RS B=72, versus +7.8/+2.0/-0.2, +5.4/+1.4/-0.3 and +3.7/+0.9/-0.2 with the primary design; the qualitative picture is unchanged.

A.7 Derivation of eq. (1)

Let $\mu_c \sim \mathcal{N}(m, \tau^2)$, $\varepsilon_c \sim \mathcal{N}(0, \sigma^2)$ independent over $c = 1..B$, $\bar{Y}_c = \mu_c + \varepsilon_c$, $\hat{c} = \arg\max_c \bar{Y}_c$. The $\bar{Y}_c$ are i.i.d. $\mathcal{N}(m, \tau^2+\sigma^2)$, so $\bar{Y}_{\hat{c}} = m + \sqrt{\tau^2+\sigma^2}\, Z_{(B)}$ with $Z_{(B)}$ the maximum of B standard normals, independent of nothing else needed. By normal-normal conjugacy $\mathbb{E}[\mu_c \mid \bar{Y}_1..\bar{Y}_B] = m + \kappa(\bar{Y}_c - m)$ with $\kappa = \tau^2/(\tau^2+\sigma^2)$, and this holds also at the random index $\hat{c}$, because $\hat{c}$ is a function of $(\bar{Y}_c)_c$. Hence $\mathbb{E}[\mu_{\hat{c}} - m] = \kappa\sqrt{\tau^2+\sigma^2}\,\mathbb{E}Z_{(B)}$ and $\mathbb{E}[\bar{Y}_{\hat{c}} - \mu_{\hat{c}}] = (1-\kappa)\sqrt{\tau^2+\sigma^2}\,\mathbb{E}Z_{(B)} = \sigma^2(\tau^2+\sigma^2)^{-1/2}\mathbb{E}Z_{(B)}$. Without the prior: for any fixed μ, $\mathbb{E}[\max_c \varepsilon_c] \leq \sigma\sqrt{2\ln B}$ and $\max_c \bar{Y}_c - \mu_{\hat{c}} \leq \max_c \varepsilon_c$, so the optimism is at most $\sigma\sqrt{2\ln B}$. For binary scores on shared items, σ^2 is the variance of the configuration-specific part of the accuracy, $\approx (1-\bar{r})p(1-p)/n$; in the simulations of Fig. 2 it is measured from the data. The formula is exact only under the stated Gaussian i.i.d. structure and ignores ties and the discreteness of accuracies.

A.8 Compute actually used

All experiments ran on a 4-core CPU quota (no GPU). Real-LLM scoring: one forward pass per (question, prompt prefix) plus short suffix passes with KV-cache reuse: 1000×72 cached score vectors

Table 5: Robustness to the shape of the candidate-quality distribution (synthetic, K=72 i.i.d. configuration effects, true SD τ=0.05 probit units, n=100, 200 draws): bias (pts) / coverage (%).

config. effects	naive	replay-boot.	hold-out	EB	worst-case	conditional	simult.
normal	+6.8/77	+2.0/94	-0.5/96	+0.2/96	-0.5/98	–/96	+6.8/98
sparse	+3.9/89	-0.3/95	-0.2/94	-4.1/74	-3.3/90	–/94	+4.0/98
heavy	+5.5/82	+0.6/92	-0.5/95	-0.9/87	-1.5/94	–/95	+5.8/98
bimodal	+6.4/80	+1.5/97	+0.2/95	+1.5/97	-0.8/98	–/97	+6.6/98

Table 6: Replay bootstrap, real pool, n=100: bias / RMSE (pts) / coverage (%) of the corrected report as a function of the number of replays b (300 benchmark draws per cell).

	b=10	25	100	300
RS B=72	+1.6 / 4.5 / 96	+1.5 / 4.7 / 95	+1.4 / 4.4 / 96	+0.9 / 4.4 / 96
MIPRO-like	+1.4 / 4.7 / 94	+0.8 / 4.3 / 96	+0.8 / 4.2 / 98	+1.1 / 4.0 / 97

for Qwen2.5-1.5B-Instruct (8-bit GGUF) took about 16 wall-clock hours on 3–4 cores, plus the Real-II grids (App. A.9) and the demonstration-leak check (about 1.5 h on one core). All statistical Monte-Carlo (synthetic: about 6 CPU-hours in total; real replays: about 1 CPU-hour) ran on the cached scores.

A.9 Real-II: a pool with true signal (model × prompt)

Pool: Qwen2.5-1.5B (from the main grid, the 8 matching configurations), Qwen2-1.5B, Qwen3-1.7B (non-thinking mode: empty think block in the assistant turn) and Qwen3-0.6B, 8-bit GGUF, each with the two groups {zero-shot, 3-shot set 1} × instructions {0,1} × cues {0,1} of the main grid, scored by the same next-token-letter rule on the first 250 questions of the item permutation (10 chunks of 25 questions; the plan of 24 chunks and a fifth model, Qwen2.5-0.5B, was cut for compute). Benchmarks of $n \in \{50, 100, 200\}$ are drawn i.i.d. from these 250 questions (with n=200 most draws contain many duplicates, so that column is least reliable), truth = accuracy over the 250. Random search with B=8 or B=K=32 and successive halving (B=32) were replayed (coordinate ascent and the MIPRO-like search use the factor structure of the main grid and are not applicable).

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Table 7: Optimism (pts) of the status-quo report with real gold labels / with permuted gold labels (all configurations then at chance, accuracy about 12%), same search procedures and benchmark draws (300 draws per cell, benchmarks drawn i.i.d. from the cached population).

	n=50	100	200	400	800
RS B=8	5.0 / 4.3	3.3 / 2.7	2.5 / 2.0	1.5 / 1.2	0.9 / 0.8
RS B=72	8.0 / 6.5	5.6 / 4.8	3.6 / 3.1	2.1 / 2.1	1.2 / 1.2
SH B=72	5.8 / 4.2	3.0 / 2.6	1.7 / 1.6	1.2 / 1.0	0.8 / 0.7
coord. asc.	7.6 / 4.9	4.7 / 3.7	2.9 / 2.4	2.0 / 1.5	1.0 / 1.0
MIPRO-like	7.6 / 5.9	4.4 / 3.5	2.3 / 2.0	1.5 / 1.1	0.7 / 0.6

Table 8: Real-II, random search over all 32 configurations: bias (pts) / RMSE / coverage (%) by reporting method.

	naive	hold-out	cross-fit	replay-boot.
n=50	+7.5/9.2/87	-1.3/11.2/96	-1.2/7.9/84	+1.5/6.1/96
n=100	+4.8/6.2/88	-0.4/8.0/97	-0.4/5.9/84	+0.8/4.5/93
n=200	+2.9/4.0/89	+0.1/5.9/96	-0.0/4.1/86	+0.5/3.2/95

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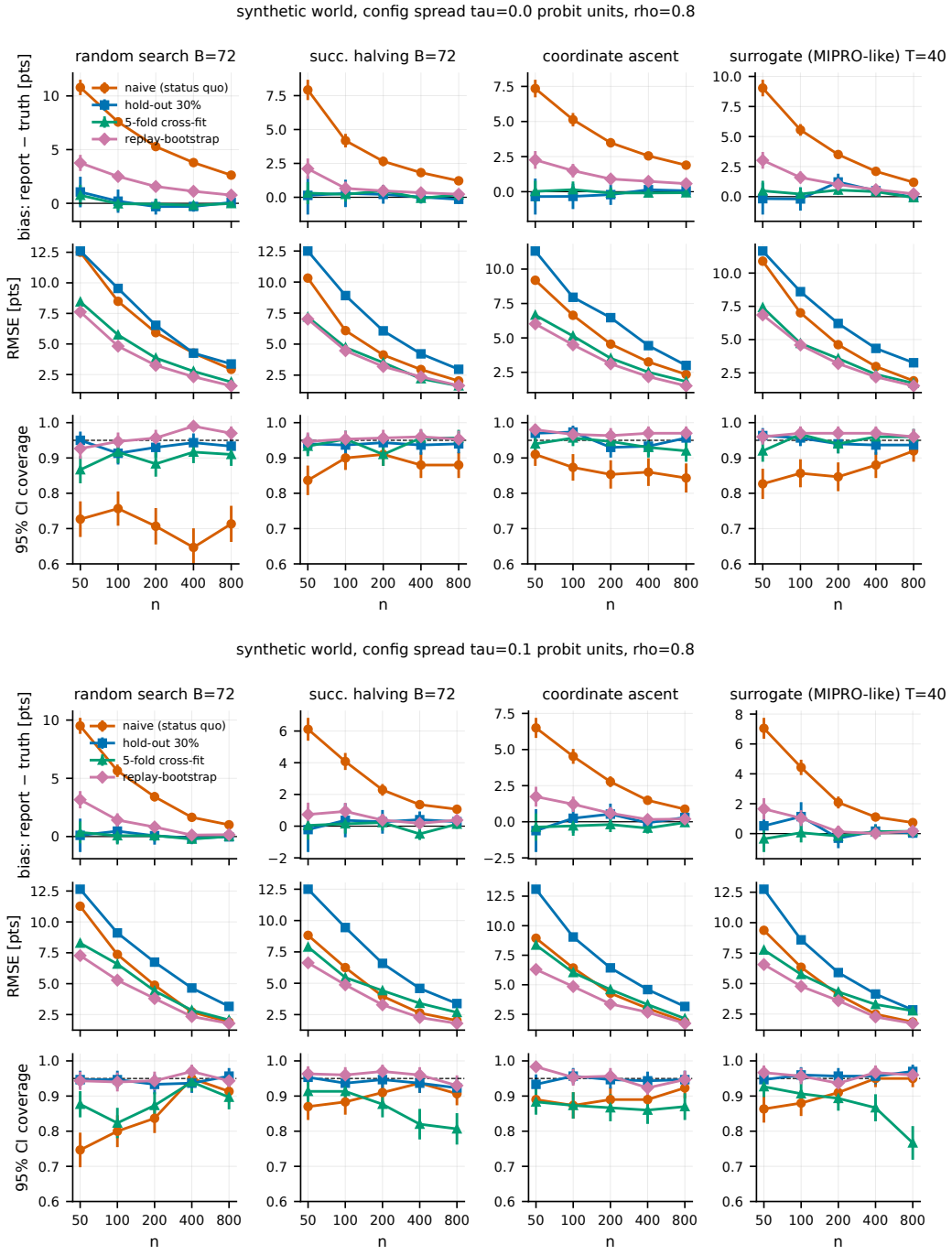


Figure 4: Synthetic counterpart of Fig. 3: configuration spread τ=0 (top) and τ=0.1 probit units (bottom), ρ=0.8, 300 draws per cell.

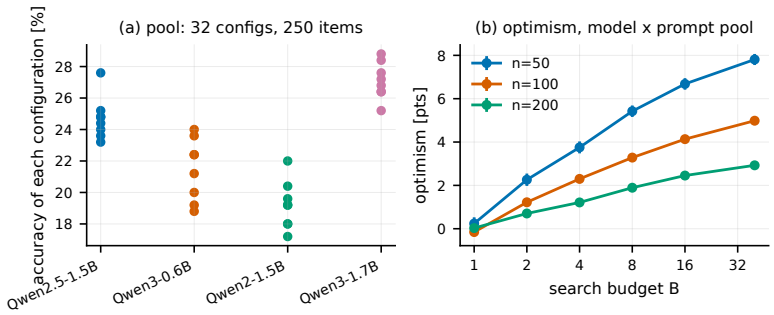


Figure 5: Real-II pool: (a) population accuracy of each configuration by model; (b) optimism of the naive report vs. search budget (400 draws per n, ± 1.96 SE).